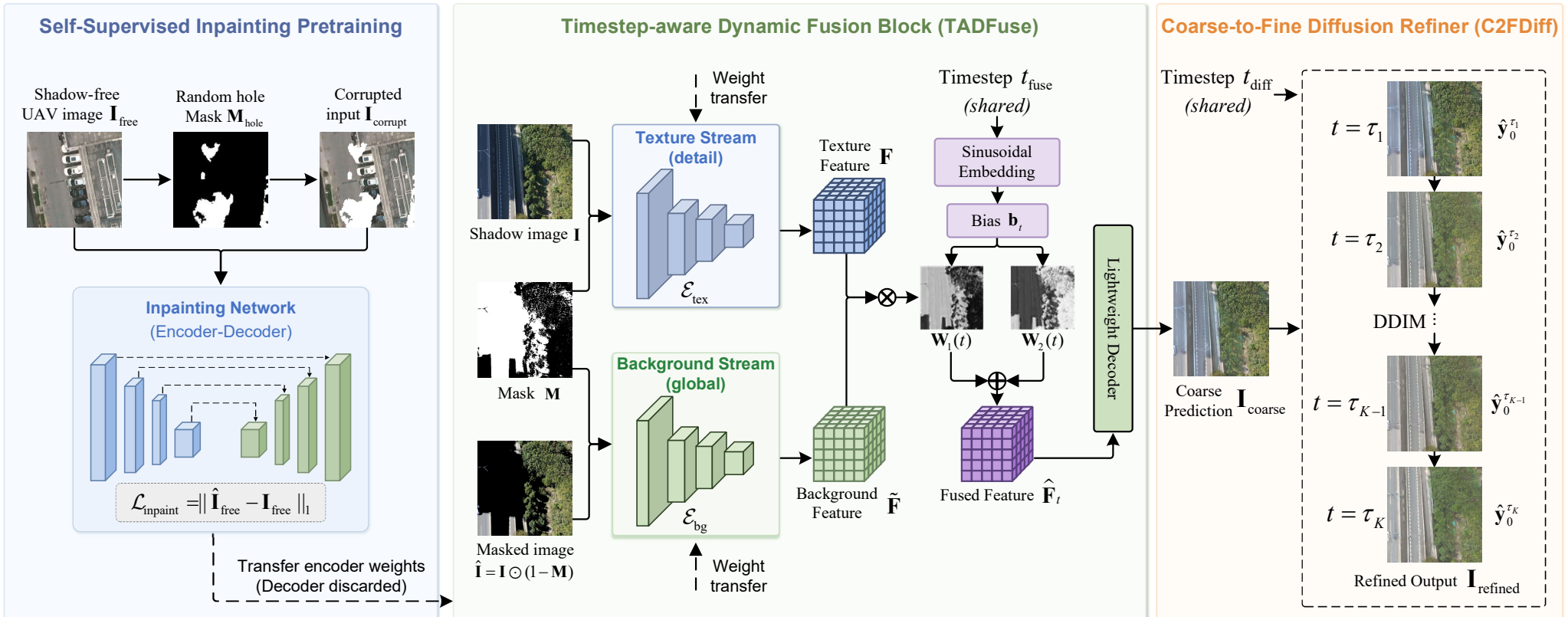

Self-Supervised Inpainting Pretraining
Shadow-free UAV image $\mathbf{I}_{\text{free}}$
Random hole Mask $\mathbf{M}_{\text{hole}}$
Corrupted input $\mathbf{I}_{\text{corrupt}}$
Inpainting Network
(Encoder-Decoder)
$\mathcal{L}_{\text{inpaint}} = \| \hat{\mathbf{I}}_{\text{free}} - \mathbf{I}_{\text{free}} \|_1$
Transfer encoder weights
(Decoder discarded)
(a)
Timestep-aware Dynamic Fusion Block (TADFuse)
Weight transfer
Shadow image $\mathbf{I}$
Mask $\mathbf{M}$
Masked image $\hat{\mathbf{I}} = \mathbf{I} \odot (1 - \mathbf{M})$
Texture Stream (detail)
$\mathcal{E}_{\text{tex}}$
Background Stream (global)
$\mathcal{E}_{\text{bg}}$
Weight transfer
Texture Feature $\mathbf{F}$
Background Feature $\tilde{\mathbf{F}}$
Timestep t_{fuse} (shared)
Sinusoidal Embedding
Bias $\mathbf{b}_t$
$\mathbf{W}_1(t)$
$\mathbf{W}_2(t)$
Fused Feature $\hat{\mathbf{F}}_t$
Lightweight Decoder
(b)
Coarse-to-Fine Diffusion Refiner (C2FDiff)
Timestep t_{diff} (shared)
Coarse Prediction $\mathbf{I}_{\text{coarse}}$
$t = \tau_1$
$t = \tau_2$
DDIM
$t = \tau_{K-1}$
$t = \tau_K$
$\hat{\mathbf{y}}_0^{\tau_1}$
$\hat{\mathbf{y}}_0^{\tau_2}$
$\hat{\mathbf{y}}_0^{\tau_{K-1}}$
$\hat{\mathbf{y}}_0^{\tau_K}$
Refined Output $\mathbf{I}_{\text{refined}}$
(c)